Chapter 10 Review

Campbell Biology, AP Edition, 12e

Chapter 10 Review

Summary of Key Concepts

Photosynthetic pigments alone cannot capture and store the energy in light. What evidence do you need to justify that claim? Could you design an experiment to prove the claim? **(BIG IDEA 2)**

Concept 10.1: Photosynthesis feeds the biosphere

- **Photosynthesis** converts the energy of sunlight into chemical energy stored in sugars.
- **Autotrophs** ("producers") produce their organic molecules from CO_2 and other inorganic raw materials; plants are *photoautotrophs* that use light energy to do this. ****Heterotrophs** (consumers, including decomposers) are unable to make their own food and live on compounds produced by others.

Using the terms producers, consumers, and decomposers, explain how photosynthesis feeds organisms, either directly or indirectly.

Concept 10.2: Photosynthesis converts light energy to the chemical energy of food

- In plants and other eukaryotic autotrophs, photosynthesis occurs in**chloroplasts**, organelles containing **thylakoids**. Stacks of thylakoids form grana. Photosynthesis is summarized as $6\ CO_2 + 12\ H_2O + \text{Light energy} \rightarrow C_6H_{12}O_6 + 6\ O_2 + 6\ H_2O$. Chloroplasts split water into hydrogen and oxygen, incorporating the electrons of hydrogen into sugar molecules. Photosynthesis is a redox process: H_2O is oxidized, and CO_2 is reduced. The **light reactions** in the thylakoid membranes split water, releasing O_2, producing ATP, and forming **NADPH**. The **Calvin cycle** in the **stroma** forms sugar from CO_2, using ATP for energy and NADPH for reducing power.

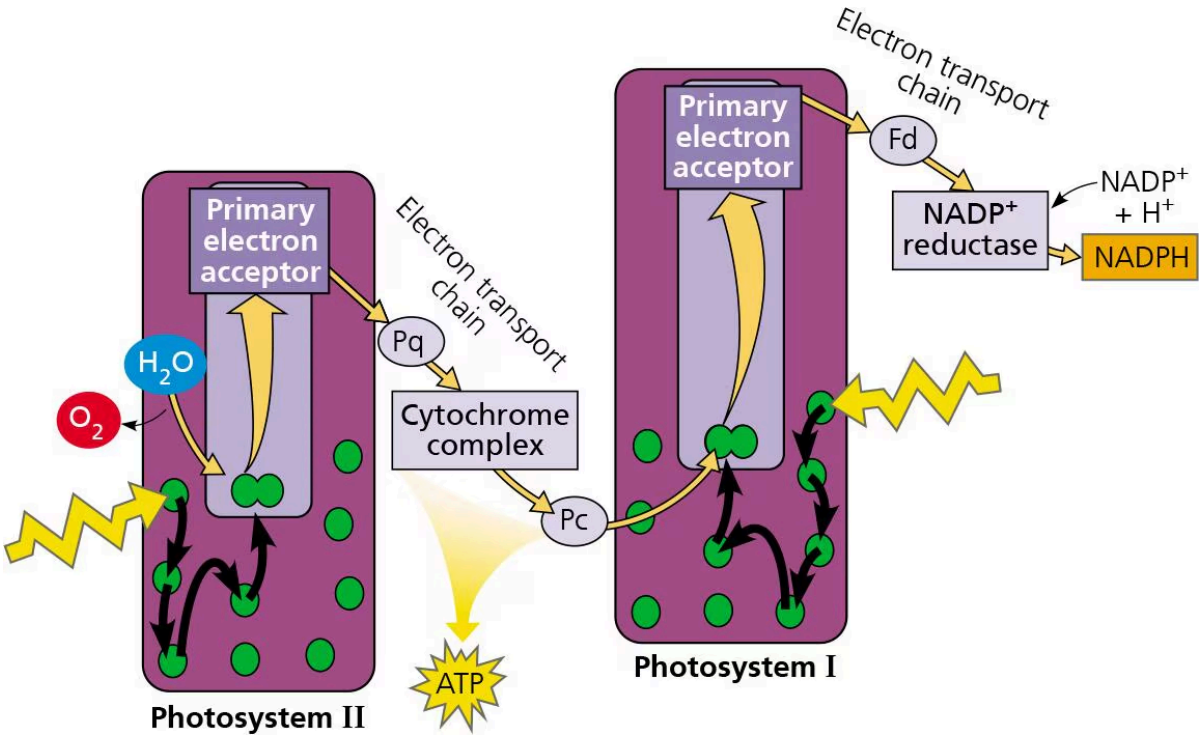
Compare the roles of CO_2 *and* H_2O *in cellular respiration and photosynthesis.*

Concept 10.3: The light reactions convert solar energy to the chemical energy of ATP and NADPH

- Light is a form of electromagnetic energy. The colors we see as **visible light** include those ****wavelengths** that drive photosynthesis. A pigment absorbs light of specific wavelengths; **chlorophyll**a is the main photosynthetic pigment in plants. Other accessory pigments absorb different wavelengths of light and pass the energy on to chlorophyll a.
- A pigment goes from a ground state to an excited state when a **photon** of light boosts one of the pigment's electrons to a higher-energy orbital. This excited state is unstable. Electrons from isolated pigments tend to fall back to the ground state, giving off heat and/or light.
- A **photosystem** is composed of a **reaction-center complex** surrounded by **light-harvesting complexes** that funnel the energy of photons to the reaction-center complex. When a special pair of reaction-center chlorophyll a molecules absorbs energy, one of its electrons is boosted to a higher energy level and transferred to the **primary electron acceptor**. **Photosystem II**

contains P680 chlorophyll a molecules in the reaction-center complex; **photosystem I** contains P700 molecules.

- **Linear electron flow** during the light reactions uses both photosystems and produces NADPH, ATP, and oxygen:



A photon of light strikes one of the pigment molecules in a light-harvesting complex of P S 2, boosting one of its electrons of a higher energy level. A primary electron acceptor relays the electron through the electron transport chain, which gives off A T P. This chain leads to P S 1, hitting P 700. More light hits pigment molecules in this system, which also leads to P 700. This excites an electron to a primary electron acceptor, which then relays it into another part of the electron transport chain where N A D P plus reductase releases N A D P H.

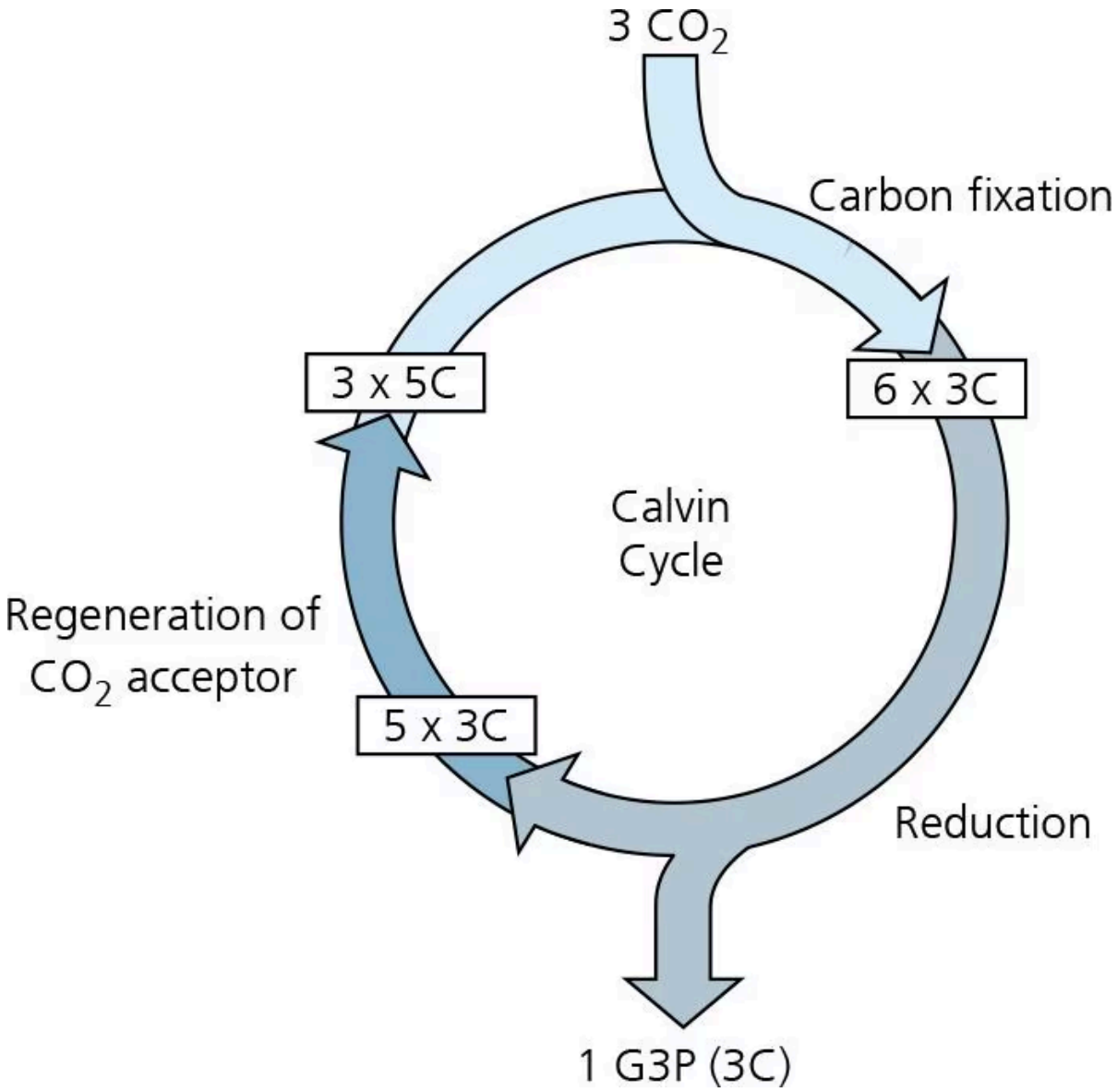
- **Cyclic electron flow** employs only one photosystem, producing ATP but no NADPH or O2.
- During chemiosmosis in both mitochondria and chloroplasts, electron transport chains generate an H+ gradient across a membrane. ATP synthase uses this proton-motive force to make ATP.

The absorption spectrum of chlorophyll a differs from the action spectrum of photosynthesis. Explain this observation.

Concept 10.4: The Calvin cycle uses the chemical energy of ATP and NADPH to reduce CO2 to sugar

- The Calvin cycle occurs in the stroma, using electrons from NADPH and energy from ATP.

- One molecule of G3P exits the cycle per three CO2 molecules fixed and is converted to glucose and other organic molecules.



3 carbon dioxide molecules enter the Calvin Cycle, they go through the process of carbon fixation to form 6 X 3 carbon molecules. After reduction, 1 G 3 P exits the cycle (taking with it 3 carbon molecules) leaving 5 X 3 carbon molecules. After regeneration of the carbon dioxide acceptor, there are 3 X 5 carbon molecules and this concludes the cycle.

DRAW IT*On the diagram above, draw where ATP and NADPH are used and where rubisco functions. Describe these steps.*

Concept 10.5: Alternative mechanisms of carbon fixation have evolved in hot, arid climates

- On dry, hot days, C3**plants** close their stomata, conserving water but keeping CO_2 out and O_2 in. Under these conditions, **photorespiration** can occur: **Rubisco** binds O_2 instead of CO_2, consuming ATP and releasing CO_2 without producing ATP or carbohydrate. Photorespiration may be an evolutionary relic, and it may play a photoprotective role.
- C4 and CAM plants are adopted to hot, dry climates.
- C4**plants** minimize the cost of photorespiration by incorporating CO_2 into four-carbon compounds in mesophyll cells. These compounds are exported to **bundle-sheath cells**, where they release carbon dioxide for use in the Calvin cycle.
- **CAM plants** open their stomata at night, incorporating CO_2 into organic acids, which are stored in mesophyll cells. During the day, the stomata close, and the CO_2 is released from the organic acids for use in the Calvin cycle.
- Organic compounds produced by photosynthesis provide the energy and building material for Earth's ecosystems.

*Why are*C4*and CAM photosynthesis more energetically expensive than*C3*photosynthesis? What climate conditions would favor*C4*and CAM plants?*

Concept 10.6: Photosynthesis is essential for life on Earth: a review

- Organic compounds produced by photosynthesis provide the energy and building material for Earth's ecosystems.

How do plants use the products of photosynthesis?